

Monsoon 2025-26 B.Tech Semester Seven CS 4033 E Quantum Computation**Total Credits:** 3 **Slot :** G (Mon 5 PM, Wed 8 AM, Thu 5 PM) **Class Room:** ELHC 401.**Instructor:** K. Murali Krishnan. (kmurali@nitc.ac.in).**Course Outcomes:** At the end of the course, the student will be able to:**CO1:** Describe the various postulates of quantum mechanics and solve elementary problems relating to their applications.**CO2:** Design elementary quantum circuits for simple requirement specifications.**CO3:** Analyze basic algorithms for Fourier transform computation, phase estimation, order finding, factoring and quantum search.

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2
CO1	H	H	H	H	-	-	-	-	-	H	-	H	H	-
CO2	H	H	H	H	-	-	-	-	-	H	-	H	H	-
CO3	H	H	H	H	-	-	-	-	-	H	-	H	H	-

About the Course: The course delivery will focus on understanding two fundamental algorithms - Schor's factoring algorithm and Grover's search algorithm. Required mathematical prerequisites, elementary facts about quantum mechanics and design of quantum gates and circuits will be developed before the algorithms and described. Although background on linear algebra will be covered in some detail, basic knowledge of linear algebra is expected. Familiarity with elementary probability theory and knowledge of elementary facts about complex numbers are prerequisites. Since quantum mechanics is a mathematical theory, students must be comfortable handling mathematics to follow the course content.

Tentative Course Plan:

Week 1 Week 2	Mathematical Foundations	Finite dimensional Hilbert spaces, Hermitian and Unitary operators,
Week 3 Week 4	Postulates of Quantum Mechanics	State space, composition, unitary evolution and measurement. Examples.
Week 5 Week 6	Quantum Gates and Circuits	Study of standard gates and elementary problems solved using quantum circuits
Week 7 Week 8	Quantum Fourier Transform (QFT)	Discrete Fourier Transform and the design of the quantum circuit for computing discrete fourier transform
Week 9 Week 10	Schor's Factoring Algorithm	Phase estimation and order finding algorithms using QFT, Schor's factoring algorithm

Week 11	Quantum Search	Grover's Algorithm
Week 12	Introduction to Quantum Communication	Introduction to quantum communication.
Week 13	Summary.	Course Summary and overview of advanced topics.

Evaluation:

At least three examinations will be conducted in the course including a Mid Test and a Final Exam. Assignments will be posted on the course page, which the students are expected to solve. However, assignments need not be submitted, and grading will be based on test based evaluations only. Weightage for the final examination will be 40% and the weightage for all the remaining examinations will be 60%. Grading will be relative.

Academic Integrity Policy.

<https://minerva.nitc.ac.in/academic-integrity>

References:

1. N. S. Yanofsky and M. A. Mannucci, Quantum Computing for Computer Scientists, Cambridge University Press, 2008.
2. M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press, 2002.
3. R. T. Perry, Quantum Computing from the Ground Up, World Scientific, 2012.